

COMPUTER ENGINEER • FULL-STACK DEVELOPER

MS. NATTHAMON KOSEM

Software Engineer (Full Stack) Backend <Python> Microservices RabbitMQ

Mobile: 099-564-9387
Email: nutkosem@gmail.com
Location: Nonthaburi / Bangkok, Thailand
GitHub: natthamon0709/portfolio

PERSONAL PROFILE

Results-driven Full-Stack Software Engineer with **4+ years of experience** engineering high-throughput backend microservices, robust RESTful APIs, and scalable distributed systems. Deep expertise in **Python (FastAPI, Flask)**, **Node.js**, and **PHP (Laravel)**, paired with **asynchronous message brokers (RabbitMQ)**, batch pipelines, and relational schema optimization (**MySQL & Oracle DB**). Hands-on mastery across all SDLC phases and automated testing (unit, functional, integration, load testing).

AREAS OF EXPERTISE

BACKEND & MICROSERVICES

Python (FastAPI / Flask) · Node.js (Express) · PHP (Laravel MVC) · Microservices Architecture · RESTful APIs (Sub-100ms) · OpenAPI / Swagger Specs

MESSAGE QUEUE & ASYNC

RabbitMQ Broker · Background Job Workers · Batch Processing · Event-Driven Architecture

DATABASES & PERSISTENCE

MySQL · Oracle Database · PL/SQL Stored Procedures · Schema Normalization · B-Tree Index Tuning · ACID Transactions

FRONTEND & WEB

Vue.js / Nuxt 4 · React & Redux · TypeScript & JS (ES6+) · Tailwind CSS / Responsive · jQuery / DOM

DEVOPS, CLOUD & TESTING

Docker Containers · Kubernetes (K8s) Ecosystem · GitLab CI/CD Automation · Unit & Load Testing · Postman / PyTest · Linux (Bash CLI)

CORE STRENGTHS

- End-to-end full-stack architecture & integration
- Database query bottleneck optimization & tuning
- Deterministic Finite State Machine workflows
- Fast technology adoption & attention to detail
- Agile / Scrum collaboration & code reviews
- High-pressure incident resolution & recovery

WORK EXPERIENCE // CAREER TRACE

Rudy Technology — 2022 – Present

Full-Stack & Backend Software Engineer (Bangkok, Thailand)

- Architected and delivered large-scale full-stack solutions and microservices connecting reactive frontends (Vue/Nuxt, React) with high-throughput backend services (**Python, Node.js, PHP/Laravel**).
- Engineered mission-critical real-time RESTful APIs, asynchronous message queue workflows (**RabbitMQ**), and background batch workers for integrating external shopping, payment, and CRM platforms with **sub-100ms latency**.
- Designed robust business workflow engines managing **order lifecycles**, **inventory synchronization states**, and transactional consistency across distributed services.
- Modeled complex relational database schemas in **MySQL & Oracle DB**, indexed query bottlenecks, and implemented ACID-compliant transactional stored procedures under strict MVC patterns.
- Expanded automated testing coverage by building comprehensive **unit tests, automated functional API tests, and performance load testing** integrated into **GitLab CI/CD**, maintaining **99.9% API uptime**.
- Collaborated with web designers to create interactive UI mockups, prototypes, and responsive dashboards using modern CSS/JS frameworks.
- Packaged microservices into **Docker** containers for cloud/Kubernetes-ready staging/production deployments, and authored contract-first OpenAPI/Swagger specifications.

Freelance Industrial Automation & HMI — 2023 – 2024

Automation & State Machine Systems Engineer

- Engineered an end-to-end deterministic **Finite State Machine** workflow coordinating Keyence barcode verification (TCP Socket), Cognex In-Sight 2800 Machine Vision (HTTP REST/TCP), and robotic dispensing motion.
- Implemented multi-protocol communication with industrial PLC over Modbus TCP (16 DI / 16 DO) with safety interlock fault-handling.

So-at Solution — 2021

Software Engineer Intern (Co-operative Training)

- Diagnosed and resolved critical web service incidents, identifying database locks and network timeouts to restore operational integrity.
- Implemented SQL stored procedures, backend controllers, and automated **Postman test suites** to validate API contracts between staging and production environments.

SCG (Siam Cement Group), Bang Sue — 2020

Engineering Intern (Student Training)

- Audited enterprise database architectures to optimize query execution plans and retrieval throughput.
- Engineered industrial IoT telemetry solutions integrating sensor hardware with backend data logging.

FEATURED PROJECTS // SYSTEMS & WORKFLOWS

Real-Time External API & Message Gateway 2023
High-throughput gateway with RabbitMQ message queues, batch workers, token rotation, and rate-limiting for external integrations.
Python • RabbitMQ • Node.js • RESTful API

Unified Core Microservices Platform 2022
Backend service endpoints handling order lifecycles, inventory sync state machines, ACID transactions, and automated load tests.
Python • Node.js • MySQL • Load Testing • CI/CD

AI Vision & NLP Matching Engines 2020–2023
Computer vision cosine similarity verification (OpenCV) and NLP intent classification pipelines in Python.
Python • OpenCV • NLP • Text Analytics

Enterprise Mobile CRM & Low-Code Web 2021–2022
Modern CRM applications with React, Redux, React Hook Form, and low-code integrations coupled with custom backend APIs.
React • Redux • TypeScript • SQL API • Low-Code

EDUCATION HISTORY

SRIPATUM UNIVERSITY — Graduated: 2021

Bachelor of Computer Engineering (B.Eng) • GPA: 3.27

- **President**, Computer and Technology Organization • **Vice President**, Student Council (Samo Sripatum)
- **National Award:** Distinguished IT Student Award for Ethics & Morality (InCIT / nCIT 2020)